

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 64



**SCIENCE
&
TECHNOLOGY-12**



In view of the advanced features, DHRUV64 is suitable for sectors such as 5G infrastructure, automotive systems, consumer electronics, industrial automation, and the Internet of Things (IoT).

SIGNIFICANT MILESTONE IN INDIA'S SEMICONDUCTOR JOURNEY

- India has achieved a significant milestone in its semiconductor journey with the launch of DHRUV64. It is a 64-bit, dual-core processor that runs at 1 GHz.
- It is a fully indigenous microprocessor developed by the Centre for Development of Advanced Computing (CDAC) under MeitY's Microprocessor Development Programme (MDP).
- DHRUV64 provides the nation a reliable, homegrown processor technology. It has enhanced multitasking capability and improved reliability. Its advanced design enables a seamless integration with a wide range of external hardware systems.
- The processor's modern fabrication leverages technologies used for high-performance chips.
- India is a major market for processors yet continues to depend on imported designs and supply chains.

WHAT DO YOU KNOW ABOUT MICROPROCESSORS?

- These are the brains of modern electronic devices, such as mobiles, computers, automobiles, medical equipment, defence systems, and satellites.
- For India, building and owning such core technologies is important to reduce reliance on foreign suppliers and support the growing electronics and digital economy.

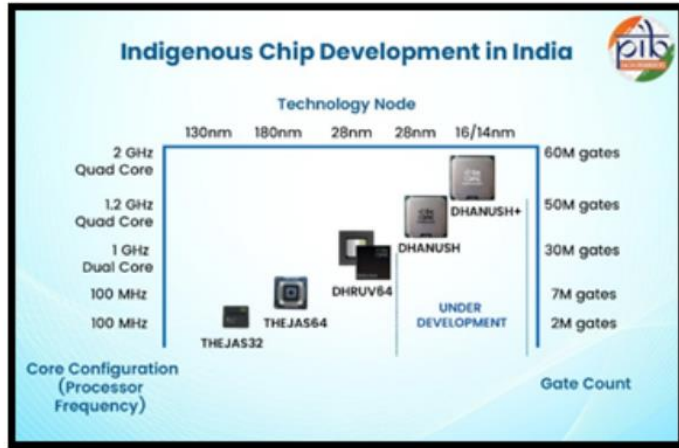
KEY DEVELOPMENTS PRIOR TO DHRUV64

- SHAKTI was designed by IIT Madras in 2018 for strategic, space and defence applications. AJIT was designed by IIT Bombay in 2018 for industrial and robotics applications.
- VIKRAM is the processor developed by ISRO-SCL in 2025 for space applications such as navigation, guidance and mission operations. It is also engineered to withstand extreme space conditions. THEJAS64 was designed for industrial automation by CDAC in 2025.





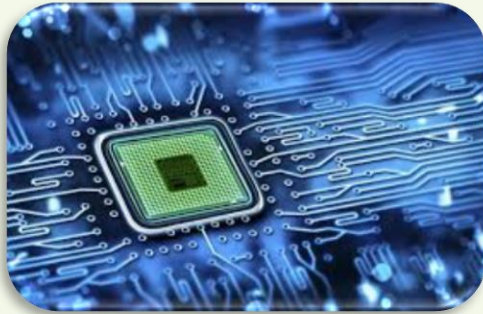
- RISC-V is an open architecture that provides a set of instructions for chip design.
- It allows wider adoption through shared innovation by industry startups and research institutions by providing common tools and standards for developers, improving collaboration across research organizations and companies.
- The Government of India launched the Digital India RISC-V (DIR-5) program to advance the vision of Atmanirbhar Bharat. It aims to establish India as a global hub for Electronics System Design and Manufacturing (ESDM).
- The initiative develops a complete portfolio of RISC-V based microprocessors. By utilizing open-source platform, DHRUV64 eliminates license costs.



- It is the third chip under the program with an overall aim to enable creation of microprocessors for the future in India.
- The first chip, THEJAS32 was fabricated at the Silterra facility in Malaysia, the second chip, THEJAS64, manufactured domestically at Semiconductor Lab, Mohali.
- The design, implementation and fabrication of DHANUSH64 and DHANUSH64+ System on a Chip (SoC) variants are currently under development.

INSTITUTIONAL ECOSYSTEM IN THE SEMICONDUCTOR JOURNEY

- MeitY plays a central role in advancing India's processor and semiconductor initiatives.
- C-DAC leads the design of India's indigenous processors. It creates processor Intellectual Properties (IPs), System-on-Chips (SoCs), and others.



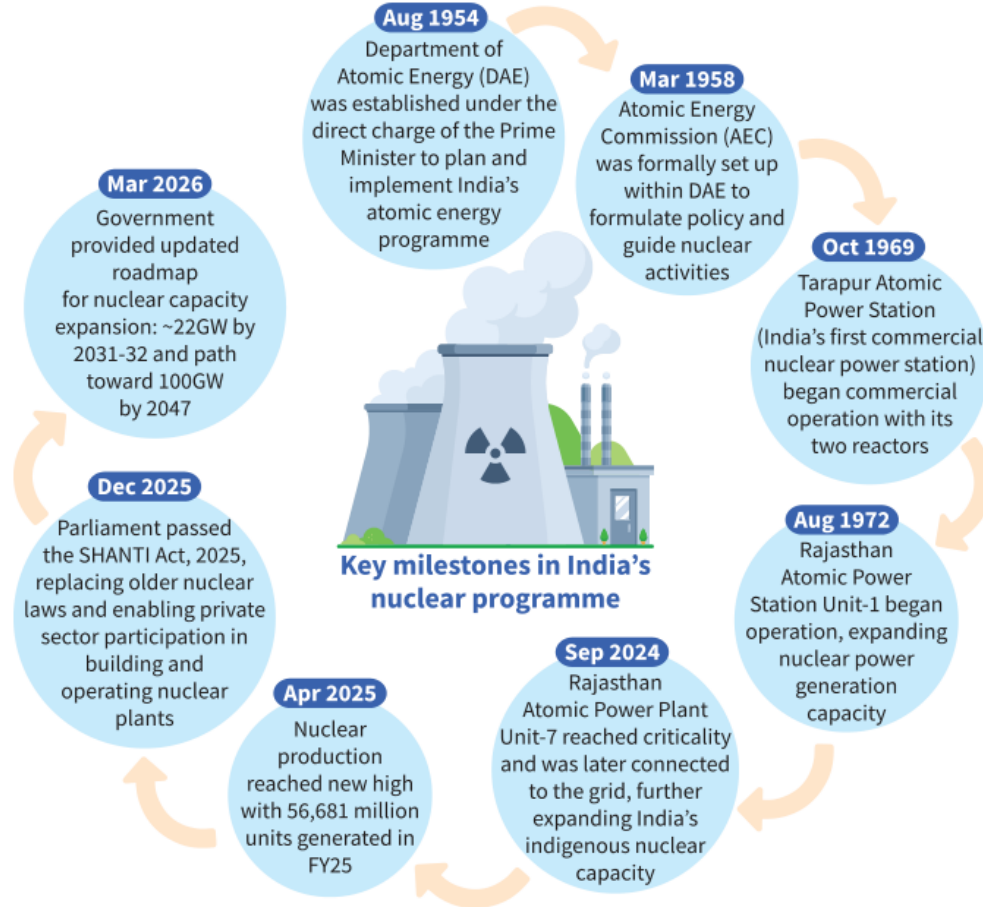
KEY NATIONAL PROGRAMMES

- India Semiconductor Mission (ISM), launched in December 2021 under MeitY, provides structured support and works with global companies to bring large semiconductor investments into the country.
- The Digital India RISC-V (DIR-V) Programme, launched in April, 2022 has played a central role in advancing India's indigenous chip design efforts.
- Design Linked Incentive (DLI) Scheme launched in 2021, aims to offer financial incentives as well as design infrastructure support across various stages of development and deployment of semiconductor design for ICs.



INDIA'S FIRST COMMERCIAL FAST BREEDER REACTOR GOES CRITICAL

- India's first commercial fast breeder reactor, the 500-megawatt **Prototype Fast Breeder Reactor (PFBR)** has gone critical - meaning a sustained **chain reaction** has begun inside the **reactor**, a step before it starts generating electricity.
- It typically **takes 12 to 18 months** for a nuclear plant to start **supplying electricity** to the grid from the date of attaining criticality .
- The **construction of PFBR began in October 2004** at **Kalpakkam near Chennai**. It was expected to be commissioned in 2011 at a cost of Rs. 3,482 crore.
- After several missed target dates, it went critical on 6th April 2026, with an expenditure of Rs. 8,181 crore.
- **BHAVINI**, the government company set up to build and run fast breeder reactors in India has plans for two more at Kalpakkam and four later at another site.
- India has about a million tons of thorium reserves, the most in the world. **Fast breeder reactors, sometimes called fast neutron reactors, convert thorium into U-233 very efficiently** because they are filled with buzzing neutrons that bombard thorium and make it U-233.



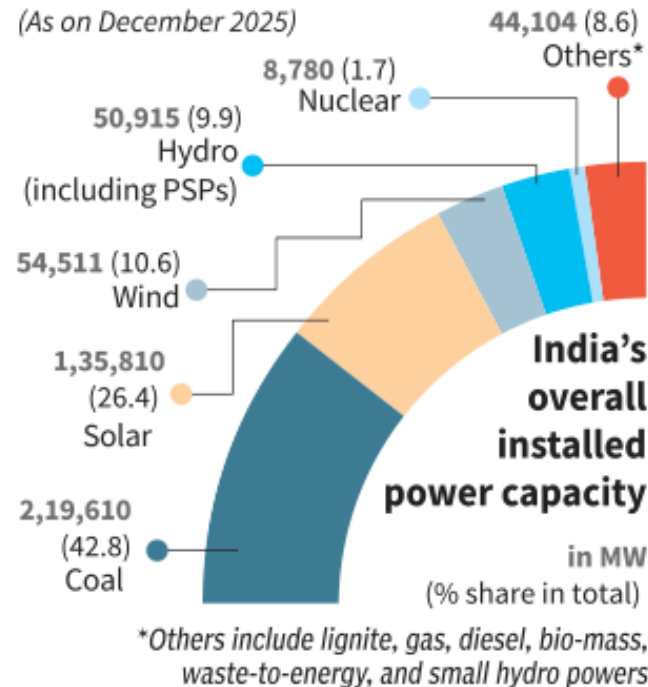
US and China lead nuclear output, India share at 2.45%

Global nuclear electricity generation among major economies

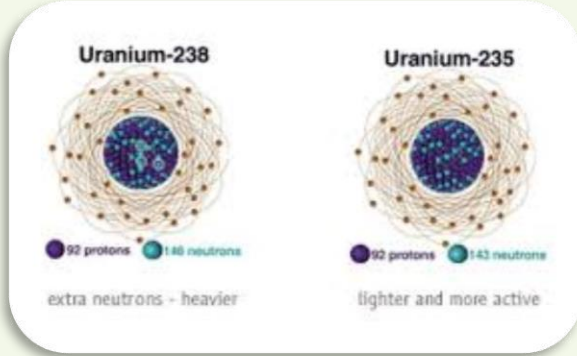
		Nuclear power supplied (GW h)	% share of total nuclear power supplied
US	●	7,81,945.1	38.4
China	●	4,17,517.6	20.5
Japan	●	84,886.9	4.2
India	●	49,909.9	2.5
UK	●	37,295.2	1.8
France	●	3,64,390.8	17.9
Russia	●	2,02,104.1	9.9
Canada	●	81,155.6	4.0
Brazil	●	14,863.0	0.7

Source: PIB, Lok Sabha, NPCIL, World Nuclear Association, Power Reactor Information System

(As on December 2025)



What do you know about U-238 in the natural environment?



- Uranium isotope (U-238) makes up over 99% of natural uranium. It is a weak radioactive, dense, and very heavy metal. It occurs naturally in the environment and can be found in almost all rock, soil, and water, including the oceans, though in low concentrations.
- External exposure to uranium is not as dangerous as exposure to other radioactive elements because the skin will block the alpha particles in the uranium. However, ingestion of high concentrations of uranium can cause health effects such as cancer of the bone or liver.
- As per USEPA, the general population is exposed to trace levels of uranium, primarily through water and food.

In India, an estimated 151 districts in 18 states have reported groundwater uranium concentration .

India powers into nuclear Stage II

India has taken a major step in its nuclear energy journey with the Prototype Fast Breeder Reactor (PFBR) at Kalpakkam achieving “criticality” on April 6, 2026. Criticality means the reactor has started a controlled and self-sustaining nuclear chain reaction for the first time, the point at which it begins producing energy. This marks India’s entry into the second stage of its long-planned three-stage nuclear programme

How the three-stage plan works

Stage I

Uranium is used to produce electricity and generate plutonium waste

Stage II

Waste plutonium (from Stage I) is used to produce even more fuel

Stage III

Finally, India plans to use its large thorium reserves to produce long-term clean energy

India’s current nuclear power landscape

Contribution	~3% of total electricity generation
Installed capacity	8.78 GW
Expansion target	22.38 GW by 2031-32 with 700 MW and 1,000 MW reactors

FLOWCHART: INDIA'S THREE-STAGE NUCLEAR POWER PROGRAMME

STAGE I: PRESSURISED HEAVY WATER REACTORS (PHWRs)

WHAT IS A BREEDER REACTOR?



A reactor that produces more fissile material than it consumes.



Neutrons from U-235 fission are slowed by heavy water (deuterium).

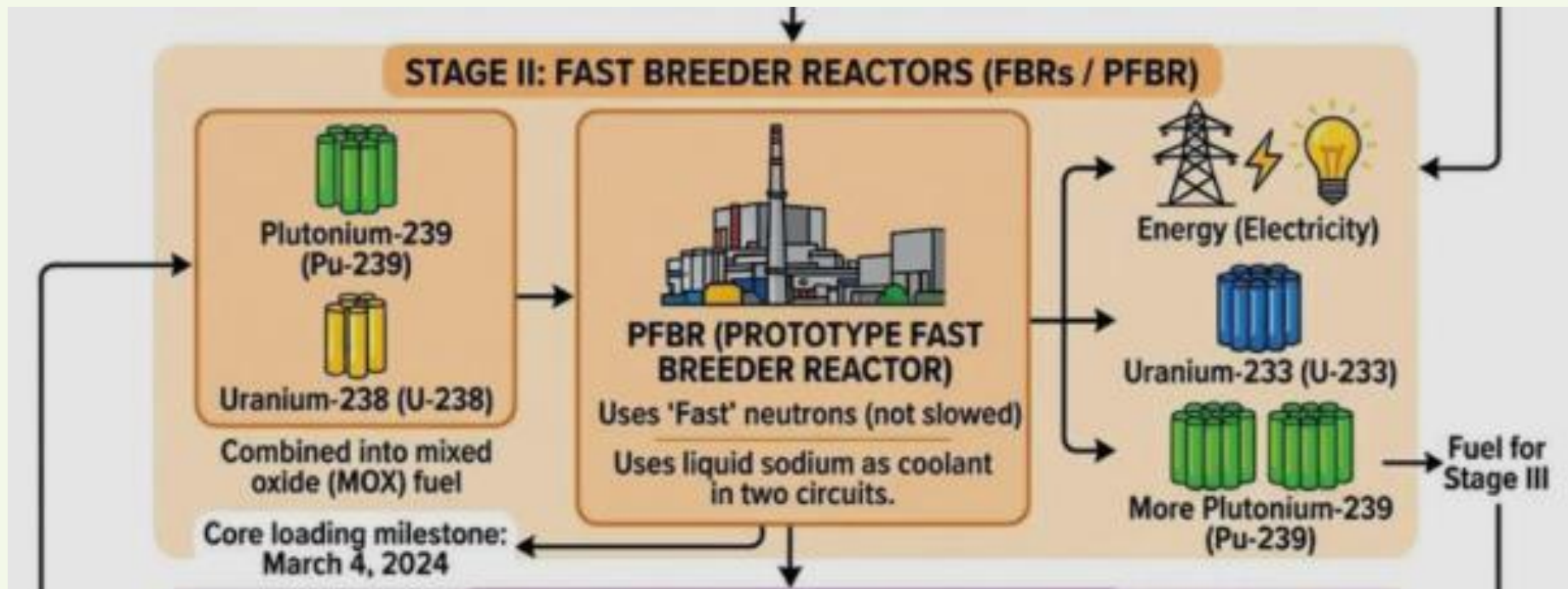


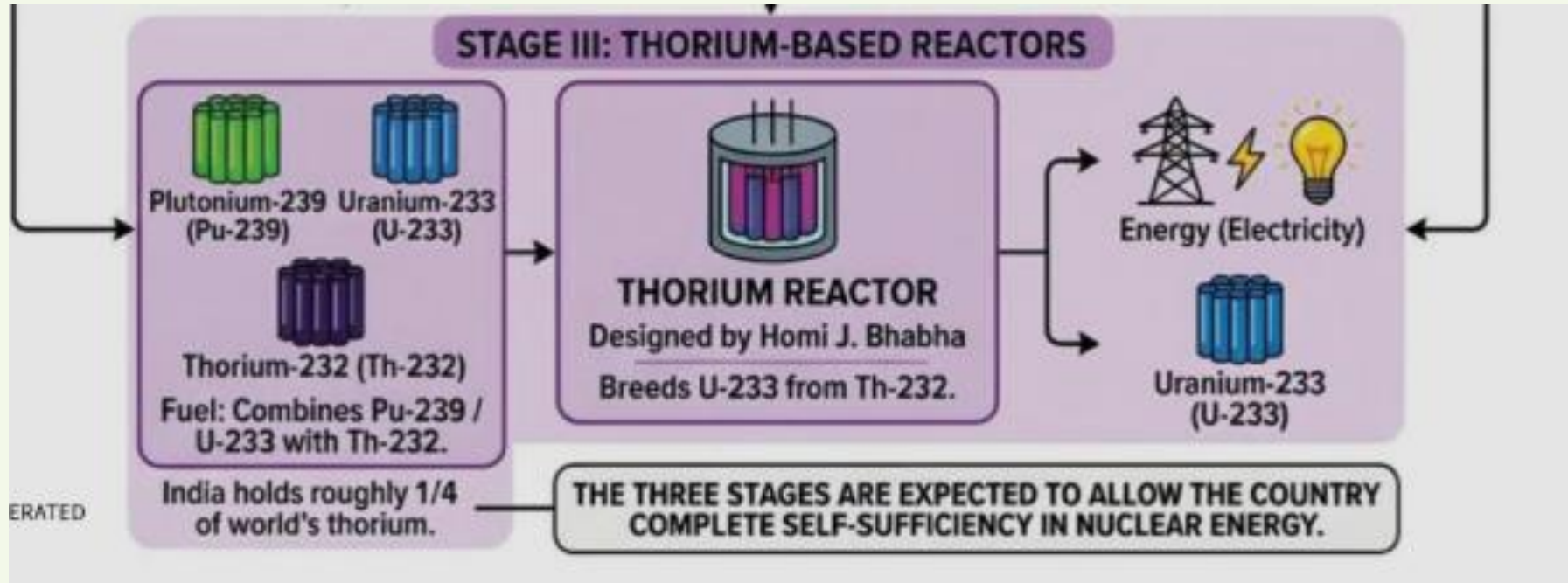
PHWR REACTOR

Undergoes Nuclear Fission
Heavy water is pressurised to prevent boiling.



Fuel for Stage II





ERATED

BUILDING FAST BREEDER REACTORS IS TECHNICALLY VERY CHALLENGING!

Learning Space



- Only Russia operates them commercially. France and Japan abandoned their efforts after finding the technical challenge unnecessary, while the US stepped back largely on economic considerations.
- China's CFR 600 is in its early stage of operation. Thus, India is on the cusp of becoming only the third country in the world to have a commercial fast breeder reactor. It will be around 12 to 18 months before the PFBR starts supplying electricity to the grid after tests and turbine synchronization.
- Former Chairman of the AEC, Anil Kakodkar pointed out that the PFBR is a very hard technology. Being the first of its kind in India, it poses several technical challenges. For example, it is a sodium-cooled reactor handling molten sodium, which reacts violently with water or air, and it is a big challenge.
- Also, the sodium needs to be kept pure, which is another challenge. Then there are materials and metallurgical issues requiring, for example, a special bespoke type of stainless steel.

THERE ARE THREE WAYS OF CONVERTING THORIUM INTO U-233!



- Fast breeder reactors do it more efficiently, but we can still put a thorium blanket around a conventional thermal reactor. It is technically possible to use a PHWR to convert the thorium into U233.
- There is also another route - called Accelerator Driven Subcritical System or ADSS under development for over 25 years. It is a machine that can generate neutrons that can be used to bombard thorium and make U-233.



- ORS stands for Oral Rehydration Salts/Oral Rehydration Solution. It is also referred to as Oral Rehydration Therapy (ORT). It is a type of fluid replacement, used to prevent and treat dehydration during various illnesses like diarrhea. Use of ORS has been estimated to decrease the risk of death from diarrhea by up to 93%.
- WHO recommends a specific quantity of each ingredient, which is important to ensure the best quality ORS reaches the public. It recommends a total osmolarity (a measure of how many dissolved particles are in a liquid) of 245 mOsm/L (milliosmoles per liter). This standard formula contains 2.6 grams of sodium chloride, 1.5 grams of potassium chloride, 2.9 grams of sodium citrate, and 13.5 grams of dextrose anhydrous (sugar per liter).

Some ORS packets produced by various organizations and sold in the market contain an approximate total sugar content of 120 grams per liter, of which 110 grams is added sugar.

What is the recent directive issued by FSSAI in the context of ORS?



- On October 15, 2025, the FSSAI issued an order banning the use of the term “ORS” on any beverage that does not adhere to the WHO’s strict medical formulae.
- As per the advisory, only medicinal ORS products that conform to standards of the WHO and are sold in pharmacies will be permitted to use that term.
- FSSAI advisory does not carry penal charges. Failing to comply with its instructions can lead to penalties under the Food Safety and Standards Act, 2006.



- A Gujarat based firm has deployed America's first ever commercial Li-Fi technology in the New York City, in collaboration with JESCO Venture Labs.
- Light Fidelity (Li-Fi) is a wireless communication technology that uses light to transmit data.
- It is entirely different from Wi-Fi. Wi-Fi mainly relies on radio waves, while Li-Fi uses visible, infrared or ultraviolet light from LED bulbs.
- Li-Fi is used for high-speed data transmission. Li-Fi offers faster speeds, greater security, and reduced interference. Since the light cannot pass through walls, Li-Fi is used to localized connections and are more secure. It offers additional control, like precise localization for asset tracking and user authentication.
- In Li-Fi system, an LED bulb which is equipped with a Li-Fi router, emits rapid pulses of light, not possible for human eye to detect. The brightness of the pulses is encoded data similar to how data is encoded in a radio wave for Wi-Fi. A Li-Fi receiver captures these signals and converts them into electronic data.

It is ideal for environments like government offices, defence establishments, airports, financial institutions and offices, where radio frequencies are restricted.

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